

The Future Is Solid — But the Battle Is at the Interface

Atomic-scale contact and pragmatic hybrid designs will decide if solid-state ever leaves the lab

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Guide: Prof. Arun Iyer · Energy Storage Centre

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Ion Transport Physics Starts at Atomic Contact

Liquids flow, conform and self-heal — solids must hold contact atom by atom.
Over to Leena Shah.



Solid-State Usually Means Hybrid, Not Pure Solid

Most so-called solid cells still need gels or liquids to keep ions moving.

Hybrid solid-state cell

A cell marketed as solid-state that pairs a solid electrolyte with gel, polymer, or liquid-assisted interfaces to maintain uninterrupted ion transport.

Oxide or sulfide ceramic with polymer interlayer to survive cracking and contact loss.



Current Flows Only Where Solids Still Touch

Liquids conform and self-heal; solids must hold atom-scale contact through vibration and cycling.

Cathode

Swells and shrinks

Expands

Contracts

Cathode Contact

Atoms must meet

Void risk

Crack risk

Solid Electrolyte

Rigid brittle path

No flow

No heal

Anode Contact

Plates unevenly

Hot spots

Dendrites

Li-Metal Anode

Moves with heat

Vibrates

Expands



Sulfides Conduct, Oxides Endure

Ion speed trades against chemical and mechanical stability at every interface.

Sulfide Electrolytes

Fast ions, fragile chemistry

Soft lattice moves Li^+ fast, reacts easily.

- Argyrodite channels give high Li^+ mobility
- Moisture control and pressure add cost

Oxide Electrolytes

Stable frame, stiff contacts

Rigid frame resists reactions, blocks ions.

- High grain-boundary resistance limits flow
- Brittle layers need flat, pressed interfaces

The framework: Conductivity favors sulfides; stability favors oxides — pick your burden.



Framework Sets the Barrier Every Ion Must Hop

Perovskites, garnets and argyrodites set Li^+ speed long before factories test survival.

Li+ Hops Via Vacancies

1

Lithium migrates vacancy to vacancy via interstitial sites; more defects mean more paths.

Sulfides Lower Barriers

2

Soft argyrodite lattice lowers hopping barrier, giving highest conductivity of the families.

Oxides Bottleneck Flow

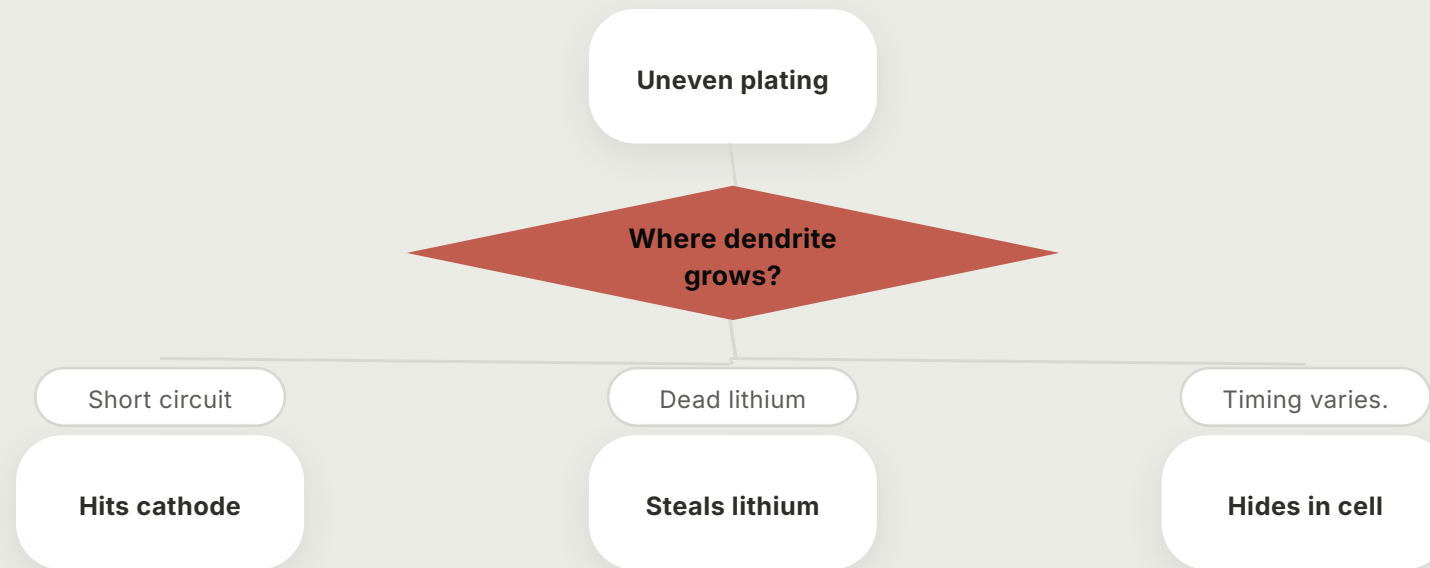
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Rigid garnet and perovskite frames endure, but grain boundaries add resistance at each crossing.

Four atomic flaws decide if millions of cells survive

Uneven Plating Splits Into Three Failures

Hot spots plate first, dendrites follow grain boundaries, and sealed cells hide growth until capacity or safety fails.



Lab Comfort Fails Gigafactory Reality

Lab success needs pressure rigs; scale demands yield, cost, and throughput without them.

BEFORE

Lab: Pressed and Pampered

Hand-built cells cycled gently under stack pressure to preserve atomic-scale contact.

AFTER

Factory: Millions, Unforgiving

Every solid electrolyte trades yield for performance

Disruption Starts Where Cost Matters Least

Low Reuse Forces a Premium-First Entry Path



Race First, Premium Next, Everyone After

Solid-state enters where performance beats price, then scales to mass market.

Phase 1: Performance

Racing cells

Performance outruns cost

Aviation entry

Power matters more than price

eVTOL demonstrators

Helicopter-like air taxis test

Phase 2: Premium EVs

High-priced segments

Performance tops competition

Premium long range

800 kilometers per charge

Phase 3: Disruption

Mass disruption

Cost and performance improve

Trucks and buses

Heavy routes electrify next

Win Premium Now to Earn the Mass Market Next

Solid-state earns scale by winning racing, aviation and premium EVs first, then driving cost down to trucks and mass EVs.

+ PROS

- Racing cells win where power beats price sensitivity
- Aviation and eVTOL demonstrators prove safety in abuse
- Premium EVs sell 800km range buyers pay extra for
- Early revenue funds dry rooms and press-free yields

– CONS

- Mass EVs need millions per month without pressure rigs
- Sulfides need dry rooms; oxides need flawless pressing

Check the Work: Sources Behind the Claims

White paper, patent data, and press record — verify the physics and market path yourself.

Hype Promises the Leap. Engineering Earns It.

Interfaces, not chemistry, decide when solid-state leaves the lab for the road.